

INA126

MICROPOWER INSTRUMENTATION AMPLIFIER

FEATURES

- LOW QUIESCENT CURRENT: 175 μ A
- WIDE SUPPLY RANGE: ± 1.35 V to ± 18 V
- LOW OFFSET VOLTAGE: 250 μ V max
- LOW OFFSET DRIFT: 3 μ V/ $^{\circ}$ C max
- LOW NOISE: 35nV/ $\sqrt{\text{HZ}}$
- LOW INPUT BIAS CURRENT: 25nA max
- 8-PIN DIP, SO-8, MSOP-8⁽¹⁾ SURFACE-MOUNT PACKAGES

APPLICATIONS

- INDUSTRIAL SENSOR AMPLIFIER:
Bridge, RTD, Thermocouple
- PHYSIOLOGICAL AMPLIFIER:
ECG, EEG, EMG
- MULTI-CHANNEL DATA ACQUISITION
- PORTABLE, BATTERY OPERATED SYSTEMS

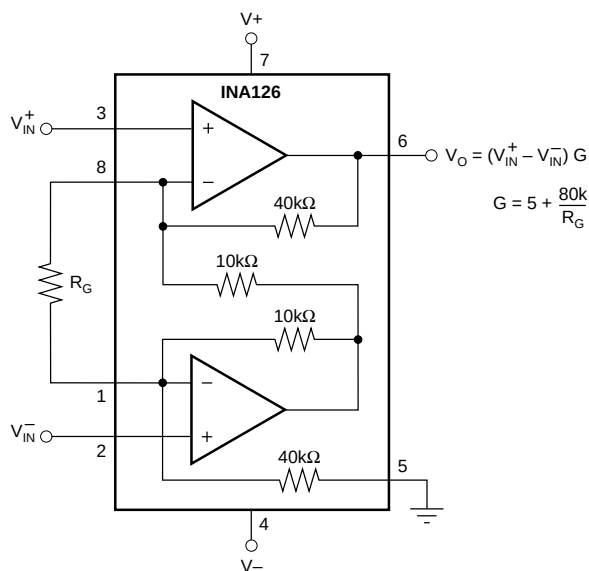
DESCRIPTION

The INA126 is a precision instrumentation amplifier for accurate, low noise differential signal acquisition. Its two-op-amp design provides excellent performance with very low quiescent current (175 μ A). This, combined with its wide operating voltage range of ± 1.35 V to ± 18 V, makes it ideal for portable instrumentation and data acquisition systems.

Gain can be set from 5V/V to 10000V/V with a single external resistor. Laser trimmed input circuitry provides low offset voltage (250 μ V max), low offset voltage drift (3 μ V/ $^{\circ}$ C max) and excellent common-mode rejection.

Package options include 8-pin plastic DIP, SO-8 surface mount, and fine-pitch MSOP-8⁽¹⁾ surface-mount. All are specified for the -40° C to $+85^{\circ}$ C industrial temperature range.

NOTE (1): Available Q1 '97.



SPECIFICATIONS

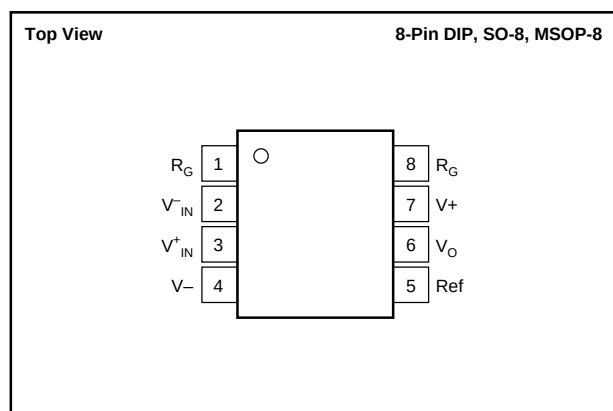
At $T_A = +25^{\circ}\text{C}$, $V_S = \pm 15\text{V}$, $R_L = 25\text{k}\Omega$, unless otherwise noted.

PARAMETER	CONDITIONS	INA126P, U			INA126PA, UA			UNITS
		MIN	TYP	MAX	MIN	TYP	MAX	
INPUT								
Offset Voltage, RTI	$V_S = \pm 1.35\text{V to } \pm 18\text{V}$		± 100	± 250		± 150	± 500	μV
vs Temperature			± 0.5	± 3		*	± 5	$\mu\text{V}/^{\circ}\text{C}$
vs Power Supply (PSRR)			5	15		*	50	$\mu\text{V/V}$
Input Impedance			$10^9 \parallel 4$			*		$\Omega \parallel \text{pF}$
Safe Input Voltage	$R_S = 0$	$(V-)-0.5$		$(V+)+0.5$	*		*	V
	$R_S = 1\text{k}\Omega$	$(V-)-10$		$(V+)+10$	*		*	V
Common-Mode Voltage Range	$V_O = 0\text{V}$	$\pm 10.5\text{V}$	$\pm 11.5\text{V}$		*	*		V
Common-Mode Rejection	$R_S = 0$, $V_{CM} = \pm 10.5\text{V}$	83	94		74	90		dB
INPUT BIAS CURRENT								
vs Temperature			-10	-25		*	-50	nA
Offset Current			± 30			*		$\text{pA}/^{\circ}\text{C}$
vs Temperature			± 0.5	± 2		*	± 5	nA
			± 10			*		$\text{pA}/^{\circ}\text{C}$
GAIN								
Gain Equation			$G = 5 \text{ to } 10\text{k}$			*		V/V
Gain Error			$G = 5 + 80\text{k}\Omega/R_G$			*		V/V
vs Temperature	$V_O = \pm 14\text{V}$, $G = 5$		± 0.02	± 0.1		*	± 0.18	%
Gain Error	$G = 5$		± 2	± 10		*	*	$\text{ppm}/^{\circ}\text{C}$
vs Temperature	$V_O = \pm 12\text{V}$, $G = 100$		± 0.2	± 0.5		*	± 1	%
Gain Error	$G = 100$		± 25	± 100		*	*	$\text{ppm}/^{\circ}\text{C}$
Nonlinearity	$G = 100$, $V_O = \pm 12\text{V}$		± 0.002	± 0.012		*	*	%
NOISE								
Voltage Noise, $f = 1\text{kHz}$			35			*		$\text{nV}/\sqrt{\text{Hz}}$
$f = 100\text{Hz}$			35			*		$\text{nV}/\sqrt{\text{Hz}}$
$f = 10\text{kHz}$			45			*		$\text{nV}/\sqrt{\text{Hz}}$
$f_B = 0.1\text{Hz to } 10\text{Hz}$			0.7			*		$\mu\text{Vp-p}$
Current Noise, $f = 1\text{kHz}$			60			*		$\text{fA}/\sqrt{\text{Hz}}$
$f_B = 0.1\text{Hz to } 10\text{Hz}$			2			*		pAp-p
OUTPUT								
Voltage, Positive	$R_L = 25\text{k}\Omega$	$(V+)-0.9$	$(V+)-0.75$		*	*		V
Negative	$R_L = 25\text{k}\Omega$	$(V-)+0.95$	$(V-)+0.8$		*	*		V
Short-Circuit Current	Short-Circuit to Ground		$+10/-5$			*		mA
Capacitive Load Drive			1000			*		pF
FREQUENCY RESPONSE								
Bandwidth, -3dB	$G = 5$		200			*		kHz
	$G = 100$		9			*		kHz
	$G = 500$		1.8			*		kHz
Slew Rate	$V_O = \pm 10\text{V}$, $G = 5$		0.4			*		$\text{V}/\mu\text{s}$
Settling Time, 0.01%	10V Step, $G = 5$		30			*		μs
	10V Step, $G = 100$		160			*		μs
	10V Step, $G = 500$		1500			*		μs
Overload Recovery	50% Input Overload		4			*		μs
POWER SUPPLY								
Voltage Range		± 1.35	± 15	± 18	*	*	*	V
Current	$I_O = 0$		± 175	± 200		*	*	μA
TEMPERATURE RANGE								
Specification Range		-40		+85	*		*	$^{\circ}\text{C}$
Operation Range		-55		+125	*		*	$^{\circ}\text{C}$
Storage Range		-55		+125	*		*	$^{\circ}\text{C}$
Thermal Resistance, θ_{JA}								
8-Pin DIP			100			*		$^{\circ}\text{C/W}$
SO-8 Surface-Mount			150			*		$^{\circ}\text{C/W}$
MSOP-8 Surface-Mount			200			*		$^{\circ}\text{C/W}$

* Specification same as INA126A, INA126U.

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PIN CONFIGURATION



ELECTROSTATIC DISCHARGE SENSITIVITY

This integrated circuit can be damaged by ESD. Burr-Brown recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

ABSOLUTE MAXIMUM RATINGS⁽¹⁾

Power Supply Voltage, V_+ to V_-	36V
Input Signal Voltage ⁽²⁾	$(V_-)-0.7$ to $(V_+)+0.7V$
Input Signal Current ⁽²⁾	10mA
Output Short Circuit	Continuous
Operating Temperature	-55°C to $+125^{\circ}\text{C}$
Storage Temperature	-55°C to $+125^{\circ}\text{C}$
Lead Temperature (soldering, 10s)	$+300^{\circ}\text{C}$

NOTES: (1) Stresses above these ratings may cause permanent damage.

(2) Input signal voltage is limited by internal diodes connected to power supplies. See text.

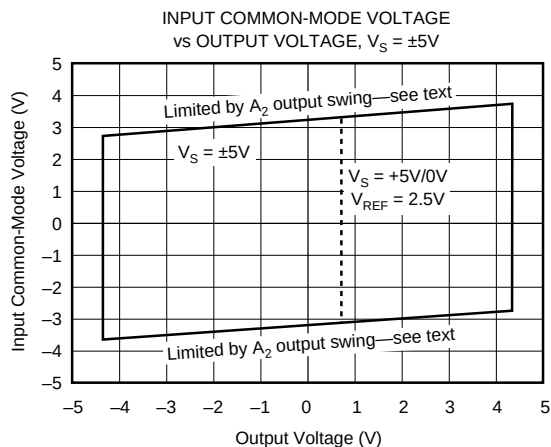
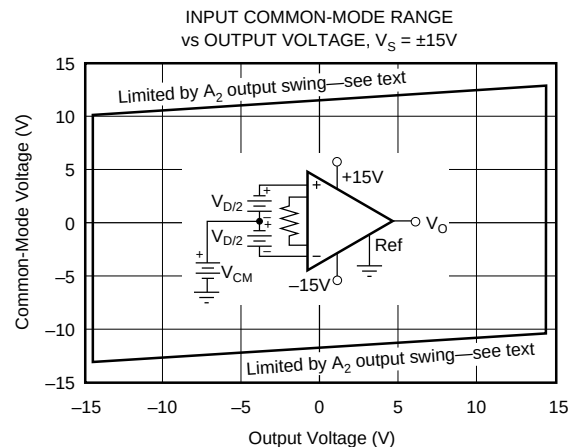
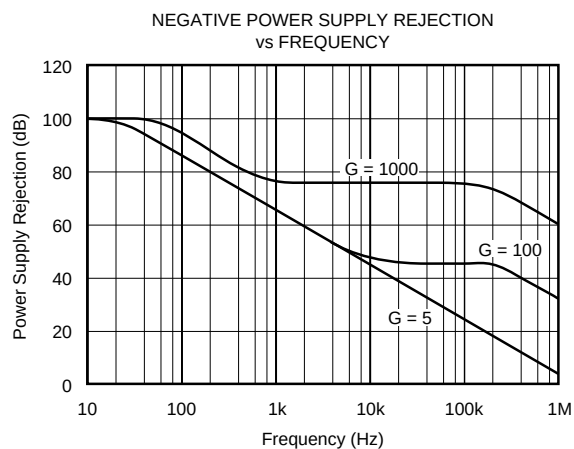
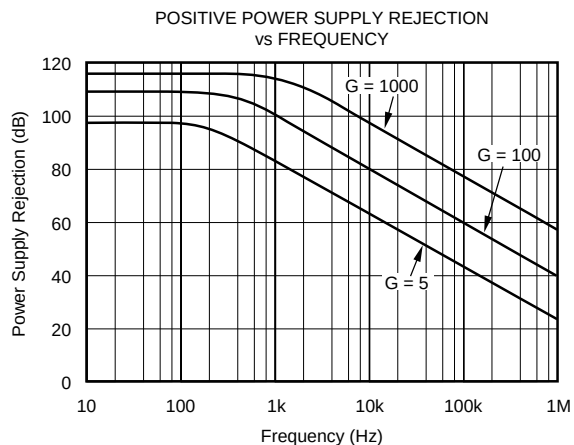
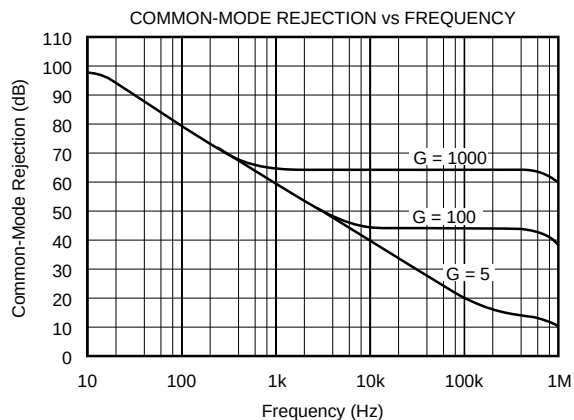
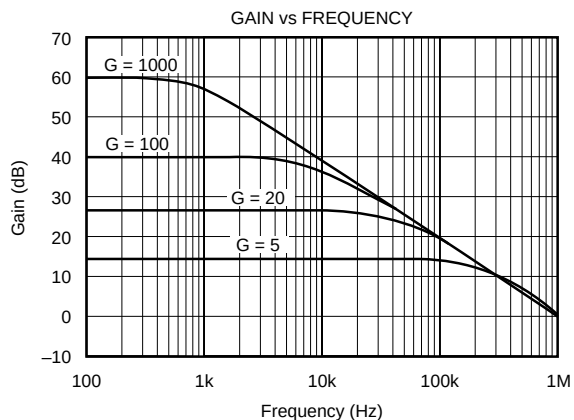
PACKAGE INFORMATION

PRODUCT	PACKAGE	PACKAGE DRAWING NUMBER ⁽¹⁾
INA126PA	8-Pin Plastic DIP	006
INA126P	8-Pin Plastic DIP	006
INA126UA	SO-8 Surface-Mount	182
INA126U	SO-8 Surface-Mount	182
INA126EA ⁽²⁾	MSOP-8 Surface-Mount	337
INA126E ⁽²⁾	MSOP-8 Surface-Mount	337

NOTES: (1) For detailed drawing and dimension table, please see end of data sheet, or Appendix C of Burr-Brown IC Data Book. (2) INA126EA and INA126E available Q1 '97.

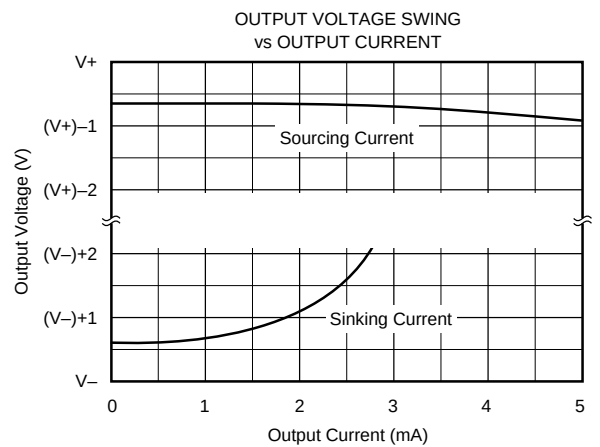
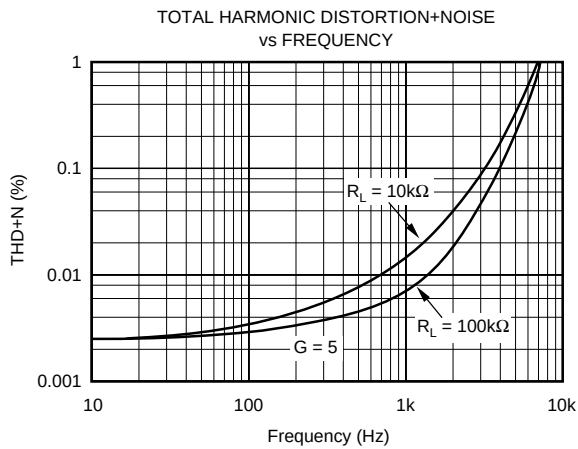
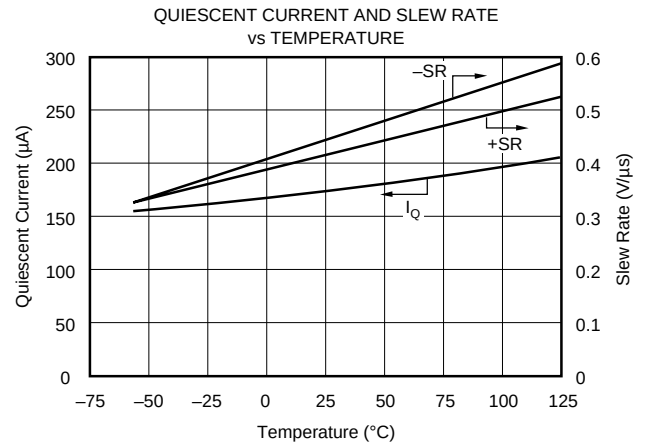
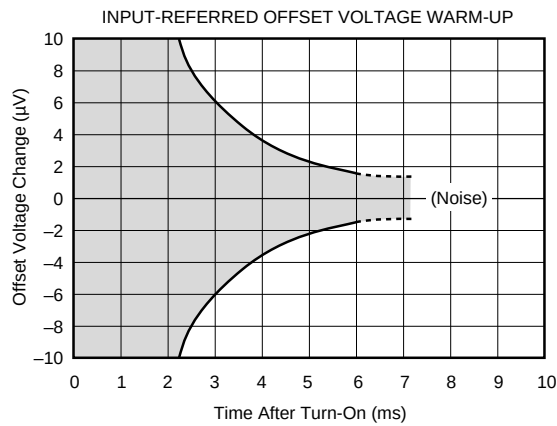
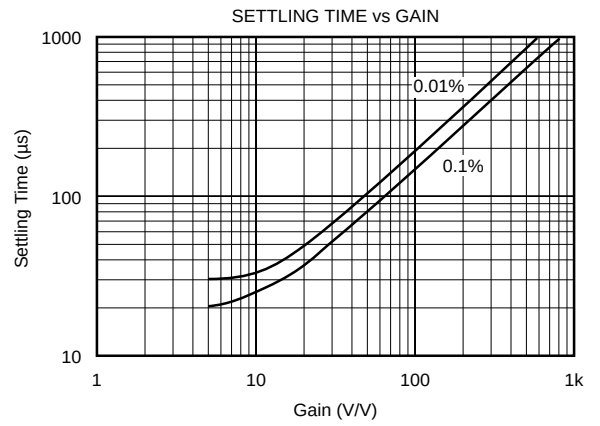
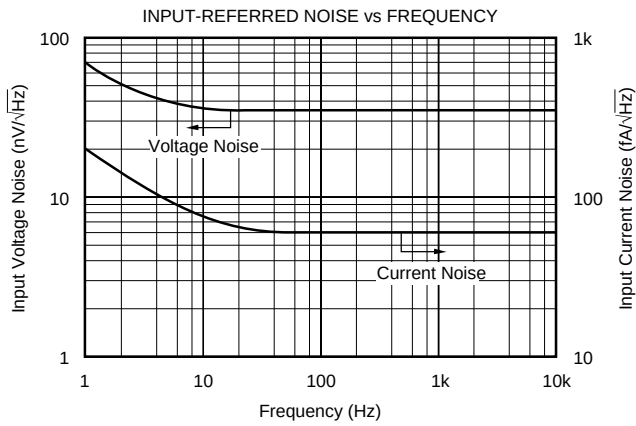
TYPICAL PERFORMANCE CURVES

At $T_A = +25^\circ\text{C}$ and $V_S = \pm 15\text{V}$, unless otherwise noted.



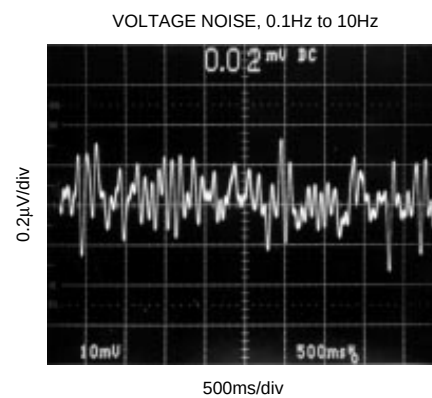
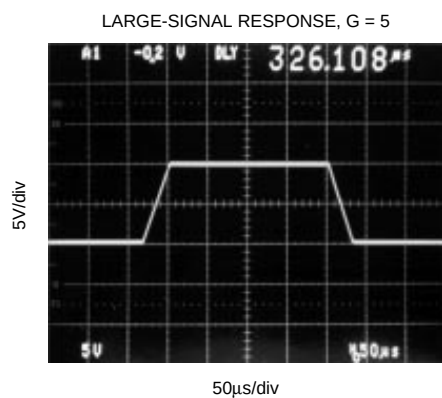
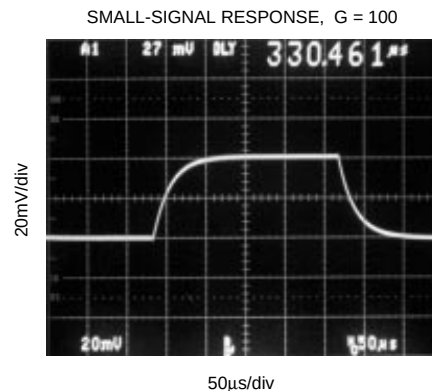
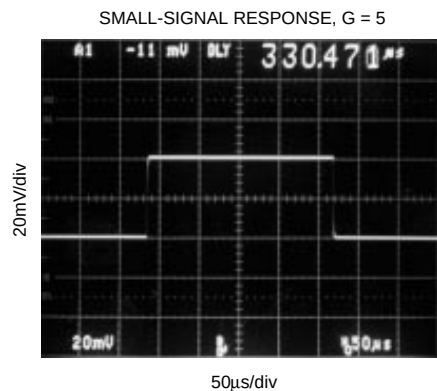
TYPICAL PERFORMANCE CURVES (CONT)

At $T_A = +25^\circ\text{C}$ and $V_S = \pm 15\text{V}$, unless otherwise noted.



TYPICAL PERFORMANCE CURVES (CONT)

At $T_A = +25^\circ\text{C}$ and $V_S = \pm 15\text{V}$, unless otherwise noted.



APPLICATION INFORMATION

Figure 1 shows the basic connections required for operation of the INA126. Applications with noisy or high impedance power supplies may require decoupling capacitors close to the device pins as shown.

The output is referred to the output reference (Ref) terminal which is normally grounded. This must be a low-impedance connection to ensure good common-mode rejection. A resistance of 8Ω in series with the Ref pin will cause a typical device to degrade to approximately 80dB CMR.

SETTING THE GAIN

Gain of the INA126 is set by connecting a single external resistor, R_G , as shown:

$$G = 5 + \frac{80\text{k}\Omega}{R_C} \quad (1)$$

Commonly used gains and R_G resistor values are shown in Figure 1.

The $80\text{k}\Omega$ term in equation 1 comes from the internal metal film resistors which are laser trimmed to accurate absolute values. The accuracy and temperature coefficient of these resistors are included in the gain accuracy and drift specifications of the INA126.

The stability and temperature drift of the external gain setting resistor, R_G , also affects gain. R_G 's contribution to gain accuracy and drift can be directly inferred from the gain equation (1). Low resistor values required for high gain can

make wiring resistance important. Sockets add to the wiring resistance, which will contribute additional gain error in gains of approximately 100 or greater.

OFFSET TRIMMING

The INA126 is laser trimmed for low offset voltage and offset voltage drift. Most applications require no external offset adjustment. Figure 2 shows an optional circuit for trimming the output offset voltage. The voltage applied to the Ref terminal is added to the output signal. An op amp buffer is used to provide low impedance at the Ref terminal to preserve good common-mode rejection.

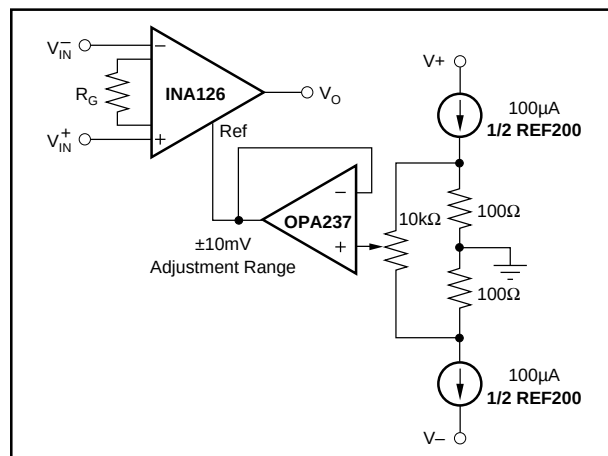


FIGURE 2. Optional Trimming of Output Offset Voltage.

DESIRED GAIN (V/V)	R _G (Ω)	NEAREST 1% R _G VALUE
5	NC	NC
10	16k	15.8k
20	5333	5360
50	1779	1780
100	842	845
200	410	412
500	162	162
1000	80.4	80.6
2000	40.1	40.2
5000	16.0	15.8
10000	8.0	7.87

NC: No Connection.

Also drawn in simplified form:

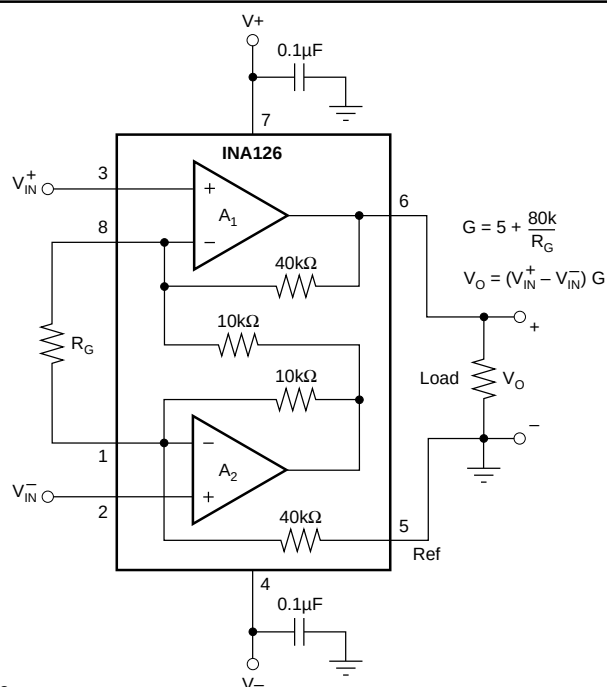
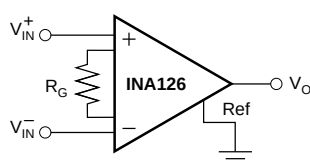


FIGURE 1. Basic Connections.

INPUT BIAS CURRENT RETURN

The input impedance of the INA126 is extremely high—approximately $10^9\Omega$. However, a path must be provided for the input bias current of both inputs. This input bias current is typically -10nA (current flows out of the input terminals). High input impedance means that this input bias current changes very little with varying input voltage.

Input circuitry must provide a path for this input bias current for proper operation. Figure 3 shows various provisions for an input bias current path. Without a bias current path, the inputs will float to a potential which exceeds the common-mode range of the INA126 and the input amplifiers will saturate.

If the differential source resistance is low, the bias current return path can be connected to one input (see the thermocouple example in Figure 3). With higher source impedance, using two equal resistors provides a balanced input with advantages of lower input offset voltage due to bias current and better high-frequency common-mode rejection.

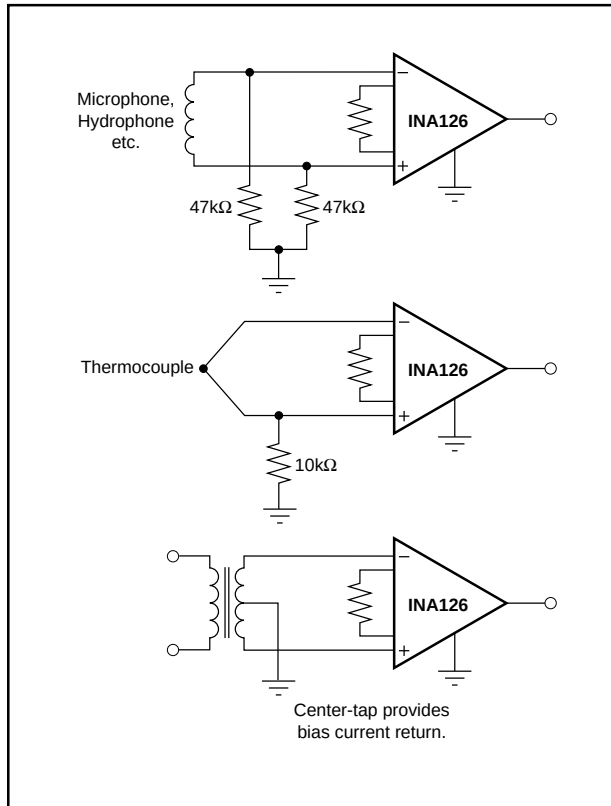


FIGURE 3. Providing an Input Common-Mode Current Path.

INPUT COMMON-MODE RANGE

The input common-mode range of the INA126 is shown in typical performance curves. The common-mode range is limited on the negative side by the output voltage swing of A_2 , an internal circuit node that cannot be measured on an external pin. The output voltage of A_2 can be expressed as:

$$V_{O2} = 1.25 V_{IN} + (V_{IN} - V_{IN}) (10\text{k}\Omega/R_G) \quad (2)$$

(Voltages referred to Ref terminal, pin 5)

The internal op amp A_2 is identical to A_1 and its output swing is limited to typically 0.7V from the supply rails. When the input common-mode range is exceeded (A_2 's output is saturated), A_1 can still be in linear operation, responding to changes in the non-inverting input voltage. The output voltage, however, will be invalid.

LOW VOLTAGE OPERATION

The INA126 can be operated on power supplies as low as $\pm 1.35\text{V}$. Performance remains excellent with power supplies ranging from $\pm 1.35\text{V}$ to $\pm 18\text{V}$. Most parameters vary only slightly throughout this supply voltage range—see typical performance curves. Operation at very low supply voltage requires careful attention to ensure that the common-mode voltage remains within its linear range. See “Input Common-Mode Voltage Range.”

The INA126 can be operated from a single power supply with careful attention to input common-mode range, output voltage swing of both op amps and the voltage applied to the Ref terminal. Figure 4 shows a bridge amplifier circuit operated from a single +5V power supply. The bridge provides an input common-mode voltage near 2.5V, with a relatively small differential voltage.

INPUT PROTECTION

The inputs of the INA126 are protected with internal diodes connected to the power supply rails. These diodes will clamp the applied signal to prevent it from exceeding the power supplies by more than approximately 0.7V. If the signal source voltage can exceed the power supplies, the source current should be limited to less than 10mA. This can generally be done with a series resistor. Some signal sources are inherently current-limited and do not require limiting resistors.

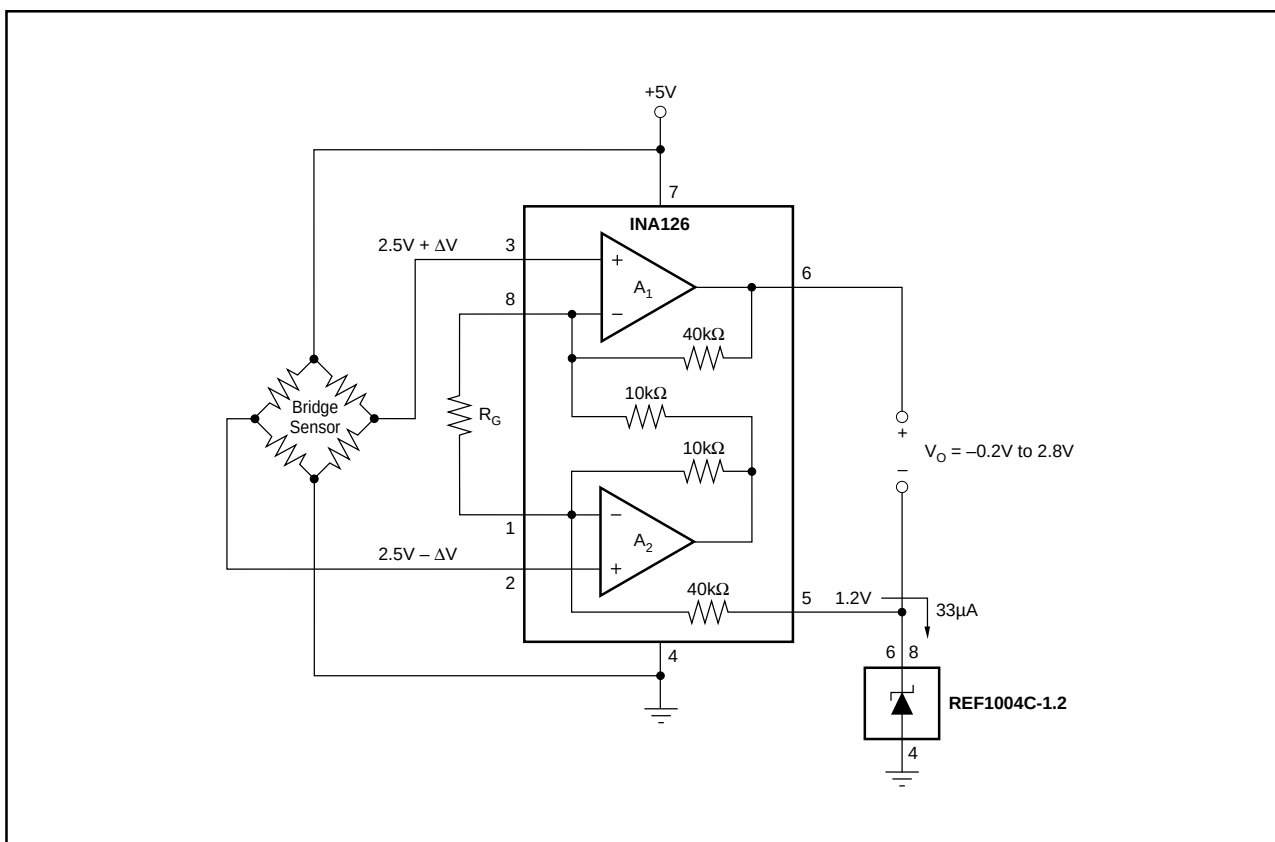


FIGURE 4. Bridge Signal Acquisition—Single 5V Supply.

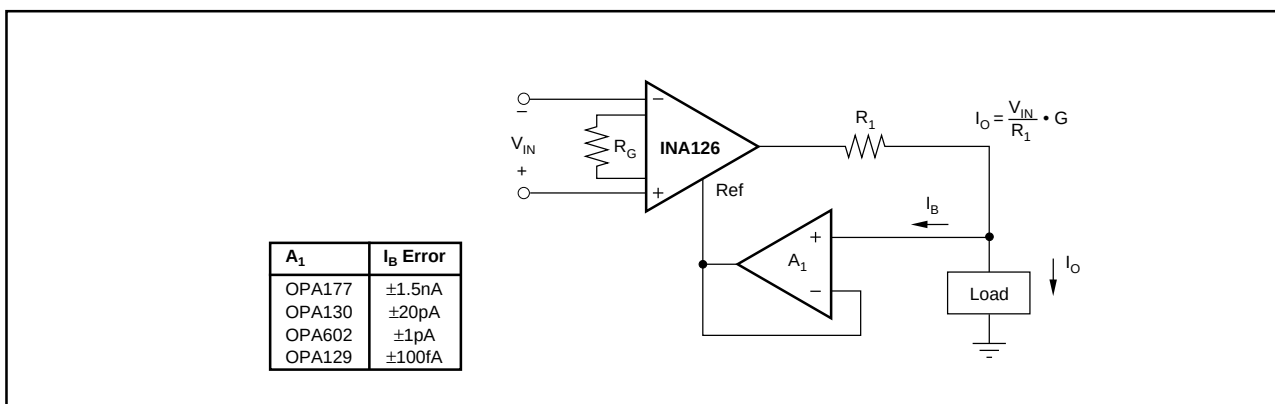


FIGURE 5. Differential Voltage-to-Current Converter.